

Session 5 — After-Session Checklist

ExInt II: Research Designs in SME Research · WU Vienna · SS 2026

Work through every step in order. Tick the box when done. Badges show where to work.

Reference project:

<https://github.com/vkiefner/sme-intl/tree/main>

Where to work:

- WRDS
- Browser
- VS Code
- Jupyter
- Terminal
- GitHub

A Identify your variables — use the WRDS Data Items sheet

VS Code

Step 1 — Retrieve the WRDS Data Items reference sheet

The WRDS Data Items sheet (distributed in Session 4) lists every standard construct, its exact Compustat field name(s), and the formula. Use it as your primary reference throughout this step.

■ **WRDS Data Items sheet — handed out in Session 4 (wrds_compustat_data_items.pdf)**

■ All field names below come directly from that sheet. The table below is a quick-reference subset — always verify against the full sheet.

VS Code

Step 2 — Choose your dependent variable (Y)

Pick one performance measure as your dependent variable. Common choices from the WRDS sheet:

Construct	Data Item(s)	Formula	Usage / Note
RoA	NI, AT	NI / AT	Most common SME performance proxy. NI = Net Income (Loss). AT = Total Assets.
RoE	NI, CSHO, PRCC_F	NI / (CSHO * PRCC_F)	Return on equity. Requires share price data — less available for private SMEs.
RoI	NI, ICAPT	NI / ICAPT	Return on invested capital. ICAPT = Invested Capital Total.
EBIT	EBIT	direct field	Operating profitability, pre-tax and pre-interest. Available as a direct field.
EBITDA	EBITDA	direct field	EBIT + Depreciation & Amortisation. Widely used in practitioner research.
Sales growth	SALE	(SALE_t - SALE_t-1) / SALE_t-1	Revenue-based performance. Requires lagging — construct in your clean script.
Tobin's Q	AT, CSHO, PRCC_F, CEQ	(AT+(CSHO*PRCC_F)-CEQ)/AT	Market-based performance. Requires stock price data.

VS Code

Step 3 — Choose your independent variable (X)

Pick the variable that captures your main theoretical concept. Examples from the WRDS sheet:

Construct	Data Item(s)	Formula	Usage / Note
R&D; Intensity	XRD, AT	XRD / AT	Innovation proxy. XRD missing → firm did not report R&D; treat as 0.
Capital Intensity	CAPX, AT	CAPX / AT	Investment in physical assets relative to firm size.
Cash Holdings	CHE, AT	CHE / AT	Liquidity proxy. CHE = Cash and Short-term Investments.
Tangibility	PPENT, AT	PPENT / AT	Share of hard assets. PPENT = Property, Plant & Equipment (net).
Foreign income share (DOI)	PIFO, SALE	PIFO / SALE	Degree of internationalization. PIFO = Pre-tax Income from Foreign Operations.

Market Value	MKVALT or CSH0, PRCC_F	CSH0 * PRCC_F	Firm market capitalisation. Use MKVALT if available, otherwise compute.
Payout Ratio	DVP, DVC, PRSTKC, IB	(DVP+DVC+PRSTKC) / IB	Shareholder distribution relative to income.

VS Code

Step 4 — Choose your control variables

Include at minimum: firm size, leverage, and firm age. Add sector (SIC/NAICS) as a categorical control if your sample spans industries.

Construct	Data Item(s)	Formula	Usage / Note
Firm Size	AT	$\log(AT)$	Log-transform reduces skewness. AT = Total Assets.
Leverage	DLTT, DLC, SEQ	$(DLTT + DLC) / SEQ$	From WRDS sheet. DLTT = Long-term Debt. DLC = Debt in Current Liabilities. SEQ = Stockholders Equity.
Firm Age	FYEAR, INCO	$FYEAR - INCO$	INCO = Year of incorporation. May be missing — use founding year if available.
Cash Flow	IBC, DP, AT	$(IBC + DP) / AT$	Operating cash generation. IBC = Income Before Extraordinary Items (CF). DP = Depreciation.
Capital Expenditure	CAPX, AT	$CAPX / AT$	Investment intensity control. From WRDS sheet: Capital Intensity.
Industry	SIC or NAICS	categorical	Use as fixed effect or dummy. SIC = 4-digit, NAICS = 6-digit.
Market-to-Book	MKVALT, BKVLP	$MKVALT / BKVLP$	Growth opportunities proxy. BKVLP = Equity Book Value per Share.
Long-term Debt	DLTT	direct field	Alternative simpler leverage measure. From WRDS sheet.
Employees	EMP	direct field (thousands)	Alternative size proxy. $emp < 0.25 \rightarrow$ fewer than 250 employees (EU SME def.)

VS Code

Step 5 — Document your variable choices in README.md

Add or update the Variables section in README.md with your chosen fields and formulas:

```
README.md

## Variables
### Dependent variable (Y)
| Construct | Data Item(s) | Formula |
|-----|-----|-----|
| RoA | NI, AT | NI / AT |
### Independent variable (X)
| Construct | Data Item(s) | Formula |
|-----|-----|-----|
| R&D; Intensity | XRD, AT | XRD / AT |
### Controls
| Construct | Data Item(s) | Formula |
|-----|-----|-----|
| Firm size | AT | log(AT) |
| Leverage | DLTT, DLC, SEQ | (DLTT+DLC) / SEQ |
| Firm age | FYEAR, INCO | FYEAR - INCO |
```

Replace with your own variables. This table becomes Table 1 in your research note.

B Verify your variables exist in WRDS

Browser

WRDS

Step 6 — Log in to WRDS and open Compustat Global

Go to wrds-www.wharton.upenn.edu → Compustat → Compustat Global Daily → Fundamentals Annual (g_funda).

Click 'Variable descriptions' to open the data dictionary. Search for each field name from your table above.

■ Confirm: the field exists, the description matches your concept, and the units are what you expect (AT is in local currency millions, EMP is in thousands).

Browser

WRDS

Step 7 — Run a quick sanity-check query on WRDS web

In the WRDS query form, pull 3 years for 5–10 firms in your target country or sector:

Date range: 2020–2022

Country: your sample (e.g. AUT, DEU, CHE)

Variables: your Y, X, and control fields

Output: screen (not download)

Check that values look reasonable — revenue in millions, RoA between -1 and 1, leverage positive.

■ If XRD (R&D;) is missing for almost all firms in your sample, you may need a proxy or a different X variable. Discuss with instructor before the pull.

C Run 01_pull_data.py — pull your data from WRDS

VS Code

Step 8 — Check your .env file

 **.env**

WRDS_USERNAME=your_wrds_username_here

■ .env must NOT be committed to Git — confirm it appears in .gitignore.

Terminal

Jupyter

Step 9 — Activate venv and run the demo notebook first

.venv\Scripts\activate

■ Mac / Linux:

source .venv/bin/activate

Open JupyterLab and run live_demo_pull.ipynb to verify your WRDS connection works on 5 firms before pulling all global data.

jupyter lab

■ If Cell 2 (connect) fails — check WRDS_USERNAME in .env and your internet connection.

Terminal

Step 10 — Run the full pull

```
python code/01_pull_data.py
```

The script pulls ALL Compustat Global variables for ALL firms, fiscal years 2015–2024, saving one parquet file per year in a new timestamped folder:

```
data/raw/
```

```
■■■ 2026-05-XX_HH-MM-SS/
```

```
■■■ column_schema.csv
```

```
■■■ pull_metadata.txt
```

```
■■■ fyear_2015.parquet
```

```
■■■ ... fyear_2024.parquet
```

■ Takes 10–30 minutes. Do not close the terminal. If it times out, re-run — the script creates a new folder each run.

VS Code

Terminal

Step 11 — Verify the output

Open pull_metadata.txt to see row counts per year. Open column_schema.csv to browse all available variables and confirm your chosen fields (NI, AT, XRD, DLT, etc.) are present.

■ data/raw/ is in .gitignore — do not commit parquet files.

D Run 02_clean.py — build your firm-year panel

Terminal

Step 12 — Run the clean script

```
python code/02_clean.py
```

Automatically finds the most recent pull folder and builds a clean firm-year panel:

```
→ reads all fyear_*.parquet chunks
```

```
→ drops duplicates and rows missing gvkey / fyear
```

```
→ converts object columns to numeric
```

```
→ sorts by gvkey, fyear
```

```
→ saves to data/processed/panel_clean.parquet
```

Jupyter

Step 13 — Explore the panel in the demo notebook

Open live_demo_clean.ipynb and run all cells. Focus on:

```
Cell 10 – missing values: are your Y and X variables well populated?
```

```
Cell 11 – panel stats: firms, years, countries
```

```
Cell 12 – describe(): do values look reasonable?
```

■ Key concern: if NI (net income) or your chosen X is missing for >50% of rows, your sample may be too thin. Note this and discuss with instructor.

Terminal

Step 14 — Quick verification in the terminal

Confirm your key variables are present and populated:

```
python -c "
import pandas as pd

df = pd.read_parquet('data/processed/panel_clean.parquet')

print(df.shape)

cols =
['gvkey', 'fyear', 'ni', 'at', 'xrd', 'dltt', 'seq', 'emp', 'sale']

print(df[[c for c in cols if c in
df.columns]].describe().round(2))

"
```

■ Adjust the column list to match your chosen variables from Step 5.

E Update README, commit scripts, push to GitHub

VS Code

Step 15 — Update README.md Data section

📄 README.md

```
## Data

| Item | Detail |
|-----|-----|
| Source | WRDS / Compustat Global |
| Table | comp_global_daily.g_funda |
| Downloaded | [today's date] |
| License | WRDS subscriber agreement |
| Fiscal years | 2015-2024 |
| Raw rows | [from pull_metadata.txt] |
| Clean rows | [from data/processed/clean_log.txt]|
```

■ Fill in the row counts. This documents the provenance of your data — a core reproducibility requirement.

Terminal

GitHub

Step 16 — Verify .gitignore, then commit and push

Confirm these are in .gitignore before running git add:

📄 .gitignore

```
data/raw/
data/processed/
*.parquet
```

Then commit only the scripts and updated README:

```
git add code/01_pull_data.py code/02_clean.py README.md
git commit -m "session 5: WRDS pull and panel cleaning scripts"
git push
```

■ Never run git add . without checking git status first when data folders exist.

Terminal

Step 17 — Confirm task all runs end-to-end

```
task all
```

Both pull and clean should run in sequence without errors. Check Taskfile.yml has:

```
tasks:
```

```
pull:
```

```
cmds: [python code/01_pull_data.py]
```

```
clean:
```

```
deps: [pull]
```

```
cmds: [python code/02_clean.py]
```

```
all:
```

```
deps: [clean]
```

F Final checklist — tick everything before Session 6

- ☐ WRDS Data Items sheet consulted for field names and formulas
- ☐ Dependent variable (Y) chosen — exact field name(s) and formula noted
- ☐ Independent variable (X) chosen — exact field name(s) and formula noted
- ☐ Control variables chosen: at minimum firm size log(AT), leverage, firm age
- ☐ Variable table added to README.md with construct, data item, and formula columns
- ☐ WRDS web interface accessed — all chosen variables verified in data dictionary
- ☐ Sanity-check query run — values look reasonable for your country/sector
- ☐ .env has WRDS_USERNAME and is NOT committed to Git
- ☐ live_demo_pull.ipynb ran successfully — WRDS connection confirmed
- ☐ 01_pull_data.py ran successfully — fyear_*.parquet files in data/raw/
- ☐ pull_metadata.txt checked — row counts per year look reasonable
- ☐ column_schema.csv opened — all chosen variable names confirmed present
- ☐ 02_clean.py ran successfully — panel_clean.parquet in data/processed/
- ☐ live_demo_clean.ipynb run — missing values checked for Y and X variables
- ☐ clean_log.txt confirms row count and column count
- ☐ README.md updated with Data section including row counts and download date
- ☐ .gitignore has data/raw/, data/processed/, *.parquet
- ☐ 01_pull_data.py and 02_clean.py committed and pushed to GitHub
- ☐ task all runs without errors from start to finish

Next session (Session 6): We build descriptive statistics and exploratory figures using 03_descriptives.py. Make sure panel_clean.parquet exists locally and your key variables (Y and X) are well-populated — check Cell 10 of live_demo_clean.ipynb. If your chosen X variable is missing for more than 50% of observations, discuss an alternative operationalization with the instructor before Session 6. Reference: github.com/vkiefner/sme-intl · Variable reference: wrds_compustat_data_items.pdf